

MINOR PROJECT

Smart Gesture-Based Communication and Health Monitoring Glove Using ESP32



Submitted by

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PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Paralyzed individuals face difficulty in **communication**

Many patients have partial hand movement but cannot speak

Existing solutions are:

1. Complex
2. Expensive
3. Not user-friendly

No integrated system for:

1. Communication
2. Health monitoring
3. Emergency alert

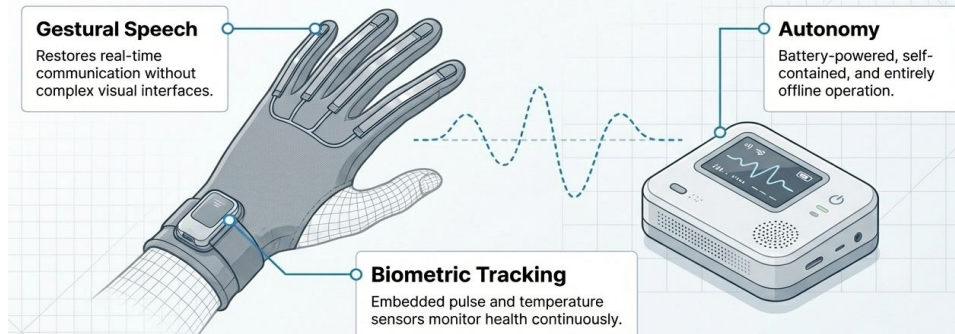
Final objective To create a **wearable assistive system** that **improves communication, monitors health, and ensures safety through real-time alerts.**

Objectives

1. Develop **gesture**-based communication system
2. Convert gestures into voice messages
3. Monitor **heart rate** and **body temperature**
4. Implement **SOS** alert using **GSM**
5. Design **low-cost** wearable device
6. Provide LCD display and buzzer alerts

A Unified Assistive Ecosystem

A single, low-cost wearable device that seamlessly translates physical movement into audible speech while continuously tracking vital signs behind the scenes.



System Architecture

Glove Unit:

- **ESP32** microcontroller
- 5 flex sensors (finger movement detection)
- Pulse sensor (heart rate monitoring)
- **Temperature** sensor (DS18B20)
- Processes sensor data
- Converts gesture into binary pattern
- Sends data via Bluetooth

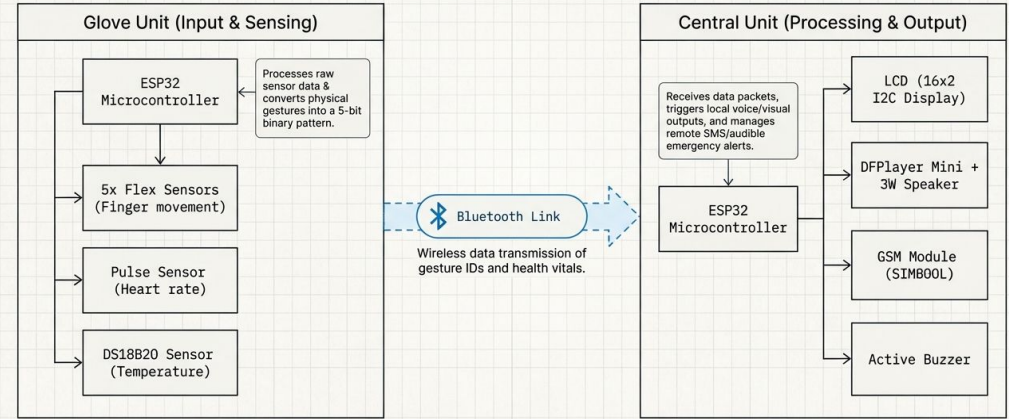
Central Unit:

- **ESP32** microcontroller
- LCD (16x2 I2C) display
- DFPlayer Mini + speaker (voice output)
- **GSM** module (SIM800L)
- Buzzer (alert system)
- Receives data via Bluetooth
- Generates output (voice + display)
- Triggers alerts and SMS

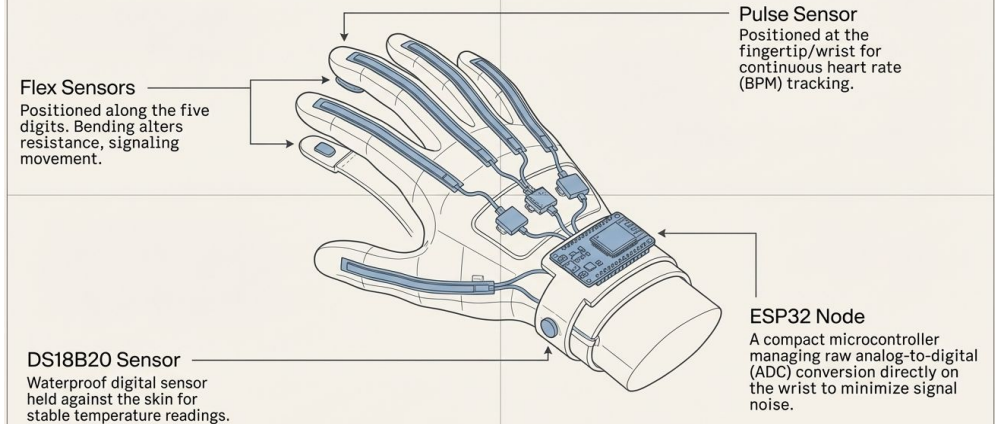
Communication:

- Bluetooth link between **two ESP32** units
- Wireless data transmission

System Architecture: The Core Engine



Anatomy of the Glove Unit

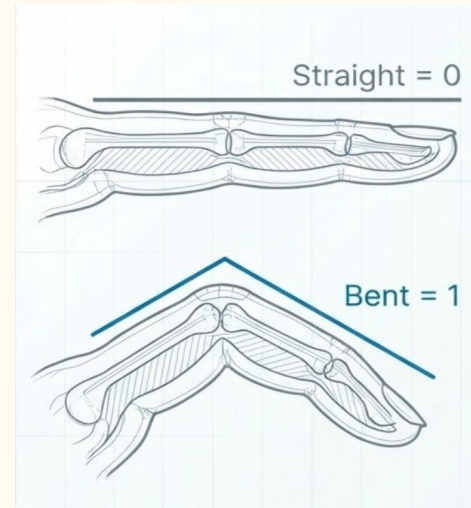


Gesture Recognition & Working

The system uses **flex sensors** to detect finger bending. When a **finger bends**, the **resistance** of the **flex sensor changes**. This change is converted into a voltage signal using a voltage divider circuit, and the **ESP32** reads it through its ADC.

The analog values are then compared with a **threshold** to determine whether a finger is **straight (0)** or **bent (1)**. These binary values from all five fingers form a gesture pattern.

Once the gesture is detected, the glove ESP32 sends the **data via Bluetooth** to the central ESP32. The **central unit processes** the data and generates the corresponding output such as voice message, LCD display, or alert.



Gesture Table for Smart Glove System

Hand Gesture Visualization & Description				Pattern	Meaning
				00000	Normal / I am OK
Open Palm	Full Fist (Hold 3 sec)	Thumb Up	Index Finger Up	11111	SOS / Emergency
				01111	Yes
				10111	No
				00111	Need Water
Thumb + Index Up	Little Finger Up	Index + Middle Up	Thumb + Index + Middle Up	11110	Washroom
				10011	Need Food
				00011	Call / Need Attention

Gesture Recognition Mechanism

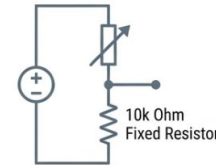
Step 1: The Physics



Straight = ~10k Ohm Bent = ~30k Ohm to 40k Ohm

Bending the flex sensor stretches the conductive ink, increasing electrical resistance non-linearly.

Step 2: The Electronics



A voltage divider circuit converts the variable resistance into measurable voltage drops.

Step 3: The Code



The ESP32's 12-bit ADC reads the voltage. Threshold logic snaps the analog reading into a clean binary state.

Straight = 0 Bent = 1

Smart Gesture Glove: Project Status & Technical Roadmap

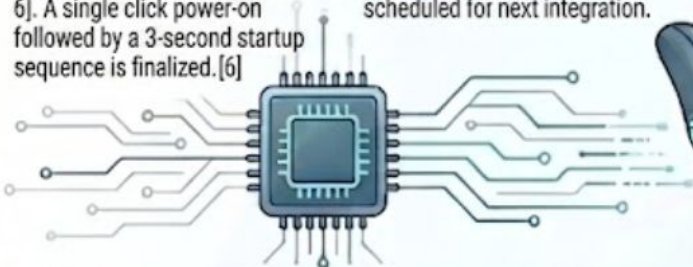
The **Smart Gesture-Based Communication Glove** uses dual ESP32 microcontrollers to translate hand movements into voice messages while monitoring vital signs. This infographic tracks the transition, highlighting how previous sensor instability has been completely resolved and all system modules are now fully integrated and operational.

System Progress & Component Status

Glove Unit: Power-On and Initialization Sequence Operational

ESP-NOW wireless communication, DS18B20 temperature sensing, and OLED UI are fully operational [6]. A single click power-on followed by a 3-second startup sequence is finalized. [6]

Central Unit: LCD receives and displays data; audio and GSM modules are scheduled for next integration.



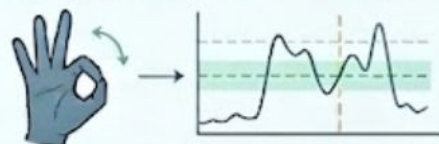
Critical System Modules: Final System Status

Module	Status	Remarks
Flex Sensors	✓ Added	Fully integrated and stabilized for functional stability.
ESP-NOW / OLED	✓ Added	Communication and UI are fully active.
GSM / DFPlayer	✓ Added	Fully integrated for voice and SOS alerts.



Technical Optimization Log

Solving Gesture Threshold Issues



"Unknown Gestures" are resolved by adjusting sensor thresholds to a specific range of 75-80.

Eliminating Signal Noise



"Wrong Detection" and "Random Bending" errors are addressed via debounce logic and signal filtering.

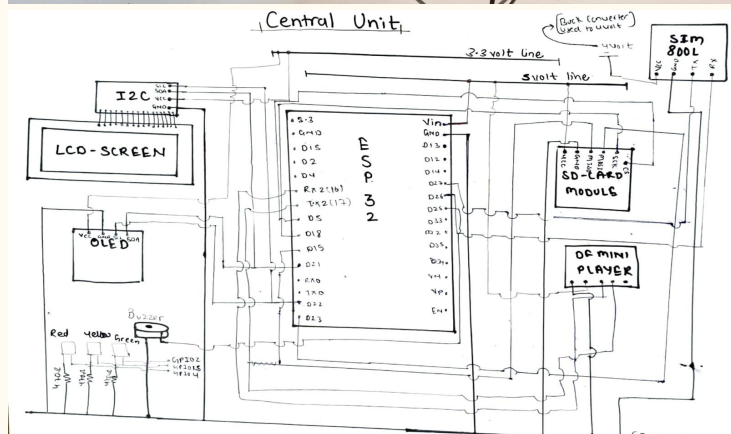
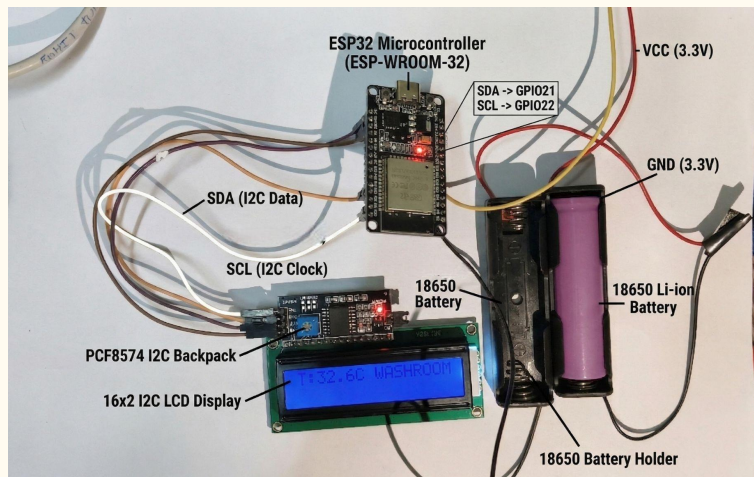
Power-On & Startup Sequence



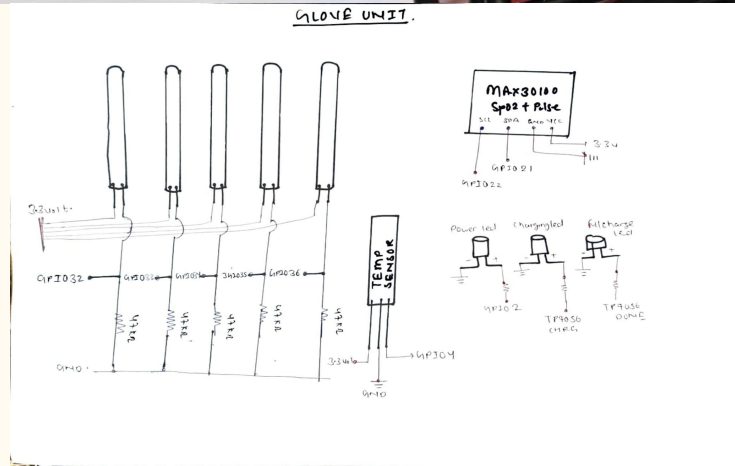
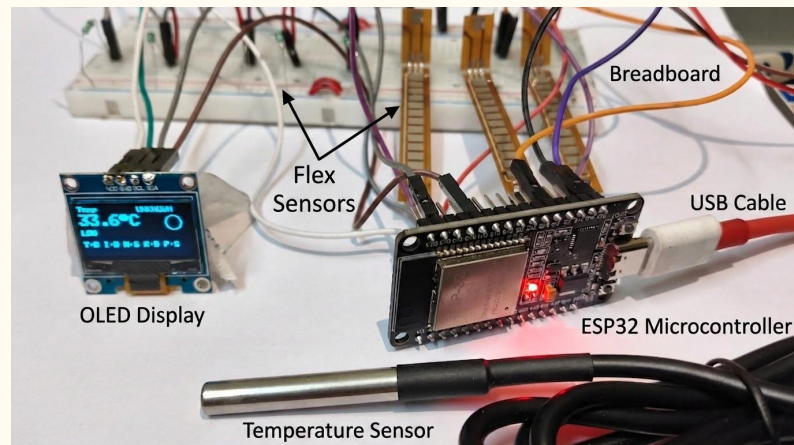
Power-On [1] → Initialize [2] (3s) → Gesture Ready [3]

BEGINNING PHASE

Central Unit Module

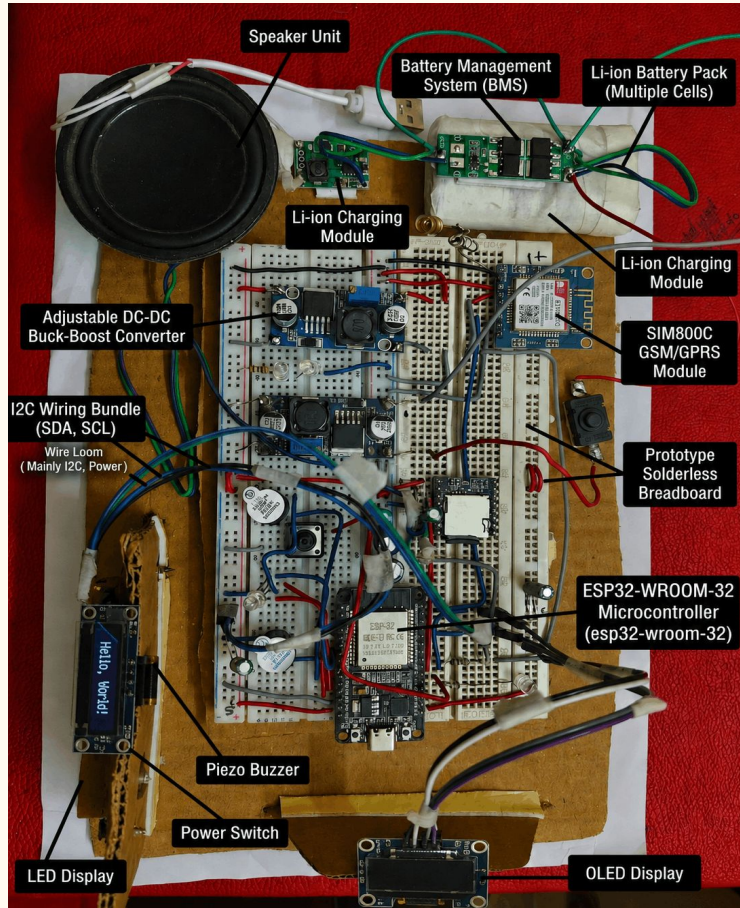


Glove Module

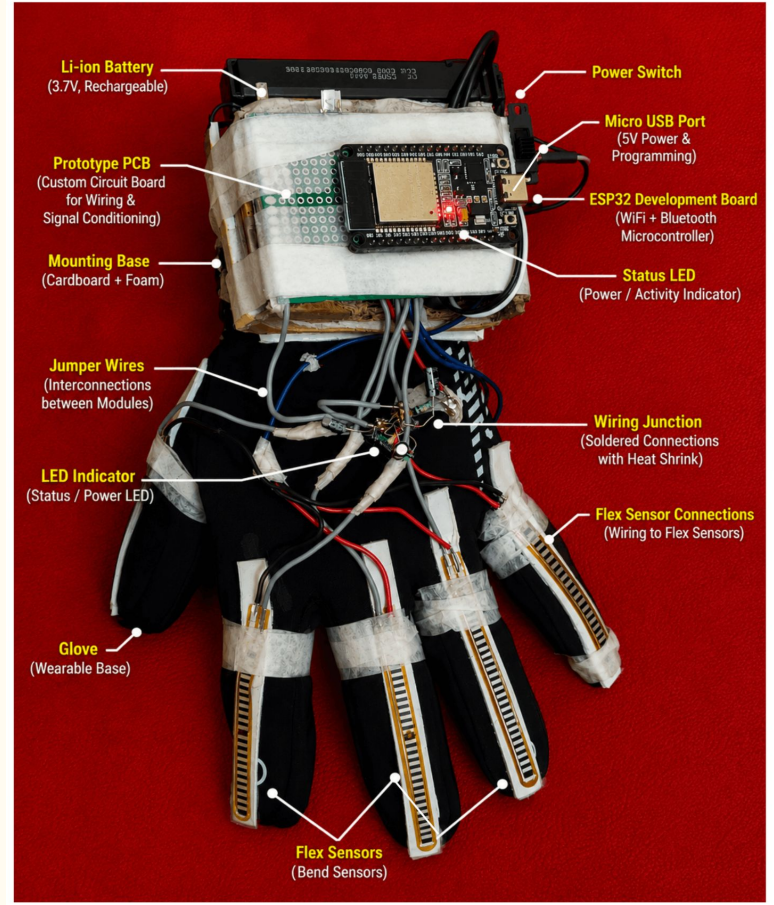


FINAL PHASE

CENTRAL UNIT MODULE

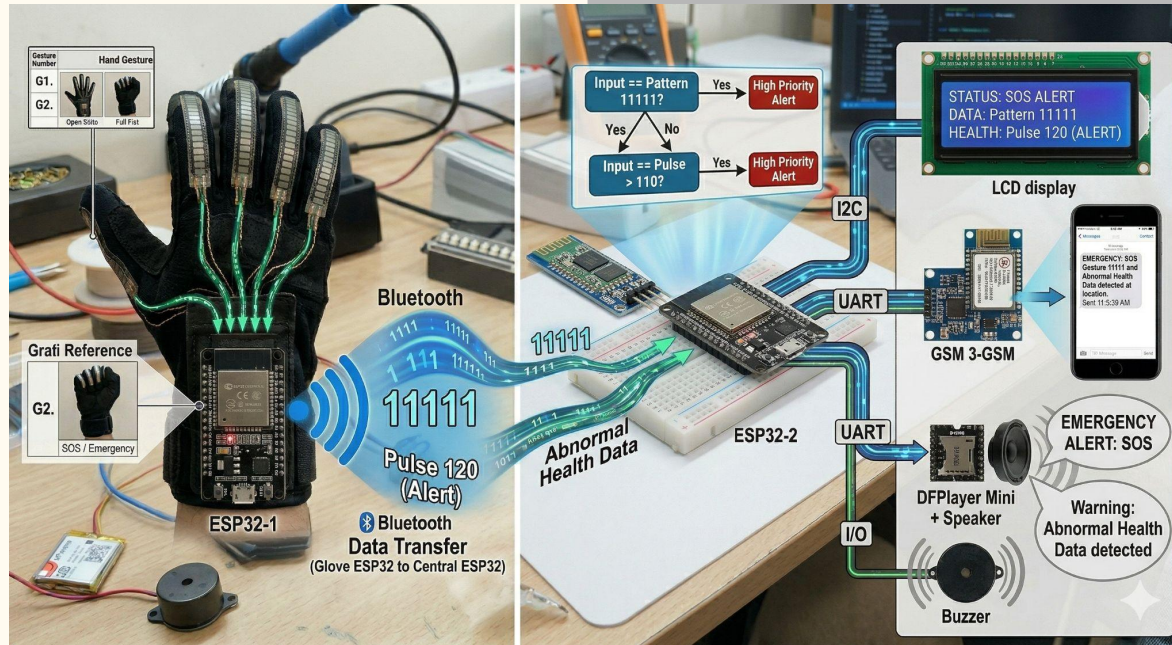


GLOVE UNIT MODULE



Communication & Output

1. Bluetooth → **Data transfer** between two ESP32
2. UART → **GSM module**, DFPlayer Mini
3. I2C → LCD display
4. Central **ESP32** processes received data
5. DFPlayer + Speaker → **Voice output**
6. LCD → **Displays message and health data**
7. Buzzer → **Alert indication**
8. **GSM module** → **Sends SMS** in emergency
9. Triggered by **SOS gesture** or abnormal values



Results, Future Scope & Conclusion

Results & Clinical Advantages

System Status Check

- ✓ Accurate 8-gesture binary detection and voice translation.
- ✓ Stable Bluetooth telemetry between nodes.
- ✓ Real-time vital display on I2C LCD.
- ✓ Reliable GSM SOS triggering on 11111 pattern.

Core Advantages



Highly Cost-Effective

~Rs 6000-7000 total build cost, dramatically cheaper than commercial AAC medical devices.



Fully Portable & Untethered

Battery-powered, decoupled design allows independent mobility without restricting the user to a bed.



Offline Resilience

GSM messaging ensures emergency reach even without WiFi or smart home infrastructure, functioning anywhere there is cellular coverage.

FUTURE SCOPE: The Next Evolution

Current: V1 Prototype



Cloud-Based Health Monitoring

Transitioning from local SD logging to secure IoT cloud databases, enabling doctors to track patient vitals remotely over time.



Mobile App Integration

Developing a dedicated application for caregivers to customize voice messages, adjust sensor thresholds, and visualize historical health data.



AI-Based Gesture Recognition

Implementing machine learning models to analyze flex sensor data, allowing for an expanded, highly personalized gesture lexicon beyond hardcoded thresholds.



Advanced Clinical Sensors

Integrating SpO2 (blood oxygen) monitoring and localized accelerometer-based fall detection to elevate the device to a comprehensive medical safety wearable.

CONCLUSION:



Enables Communication

Restores a critical voice to paralyzed individuals, bypassing the fatigue of writing and the complexity of mobile apps through natural, intuitive hand gestures.



Real-Time Health Monitoring

Provides continuous, localized biometric tracking (Heart Rate & Temperature) to ensure baseline vitals are always visible to nearby caregivers.



Ensures Safety Through Alerts

Eliminates the vulnerability of isolation. Priority-based SOS triggers guarantee that remote caregivers are instantly notified during medical emergencies.



Low-Cost, Portable Wearable

Achieves a highly modular, battery-powered wearable form factor using affordable, off-the-shelf ESP32 architecture, making assistive technology financially accessible.

MICROCONTROLLERS & SOFTWARE FRAMEWORKS



Espressif Systems, "ESP32 Series Datasheet," 2023.
Primary technical specification for the dual-core microcontrollers managing the glove and central units.



Espressif Systems, "ESP-NOW User Guide," 2023.
Documentation for the low-latency, connectionless wireless protocol used for peer-to-peer data transfer.



Arduino, "Arduino IDE and ESP32 Documentation," 2024.

Development environment and board packages used to write and upload the system firmware.



NXP Semiconductors, "I2C-bus Specification and User Manual," n.d.
Standard communication protocol used to interface the LCD and health sensors with the ESP32.

COMMUNICATION MODULES & POWER DESIGN



SIMCom, "SIM800L GSM Module Specification," 2020.
Technical requirements for cellular module triggering emergency SOS SMS alerts.



Texas Instruments, "Power Supply Design Considerations for GSM/GPRS," 2020.

Critical application note for handling high-current bursts required by the SIM800L module.



NanJing Top Power ASIC Corp., "TP4056 1A Standalone Linear Li-Ion Battery Charger Datasheet," n.d.
Specifications for charging circuit managing the glove unit's portable power.

SENSOR HARDWARE & AUDIO COMPONENTS



Spectra Symbol, "Flex Sensor FS-L-00SS/FS-L-0095 Datasheet," 2020.
Technical specs for the sensors that detect finger bending via resistance changes.



Analog Devices/Maxim, "MAX30102 Pulse Oximeter and Heart-Rate Sensor Datasheet," 2021.

Core reference for high-sensitivity optical sensor for heart rate and SpO2 tracking.



Maxim Integrated, "MAX30100 Pulse Oximeter and Heart Rate Sensor Datasheet," 2021.

Secondary reference for integrated pulse oximetry sensor design.



DFRobot, "DFPlayer Mini Datasheet," 2022.
Guide for the hardware module responsible for playing MP3-based voice messages.

SCHOLARLY RESEARCH & HEALTH STANDARDS



Sharma, A., and Gupta, R., "Low-cost assistive communication device," 2021.
Academic research published in the Indian Journal of Embedded Systems regarding IoT-based devices for paralyzed patients.



M. T. Pichaiak et al., "Open-Source IoT-Based Remote Health Monitoring," 2023.

Study from the Journal of Low Power Electronics on using ESP32 for remote health tracking.



World Health Organization, "Assistive Technology Fact Sheet," 2024.
Global health standards and requirements for assistive devices to improve patient quality of life.